Guangzhi Chen

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EDUCATION

The University of Texas at Austin **August 2024 - December 2027**
Bachelor of Science, Aerospace Engineering **Cumulative GPA:** 3.91/4.00
Relevant Courses: Linear System Analysis, Feedback Control System, Spacecraft Dynamics, Flight Dynamics, Spacecraft Systems Design, Senior Design

EXPERIENCE

Software/GNC Engineer (Co-founder), **Texas Ascent & Landing** **July 2026 - Present**

- Co-founded a student team developing an electric propulsive-landing hopper; building a reusable vehicle simulation in NASA's Trick framework with JEOD, structured as a general core with vehicle-specific configurations
- Unit-testing 6-DOF rigid-body dynamics and writing shell-script tests that check the sim against physical invariants (energy and momentum balance) & performance benchmarks, as groundwork for a CI/CD pipeline
- Conducting system-level design and trade studies across vehicle configuration, experimental objectives, and navigation architecture

GNC Research Intern, **CDUS Lab (Dr. Maruthi Akella) - USSF SSTI** **June 2026 - July 2026**

- Cut spacecraft position RMSE 7× (0.132 m → 0.020 m, below the 0.138 m noise floor) by rebuilding an MEKF's translation model around 6 Relative Orbital Elements with an exact Keplerian state-transition matrix
- Validated in Python on Stanford SLAB's SHIRT dataset (2,371 frames, 2 orbits of J2-perturbed truth)
- Derived the ROE-to-camera measurement map and its 3×6 Jacobian, and resolved the 3-measurement/6-state rank deficiency at initialization with a diffuse-prior, minimum-norm seed

Activities

Controls Engineer, **Quadcopter Controls Simulation (Personal Project)** **August 2025 - May 2026**

- Built a 6-DOF quadcopter simulator and flight controller from scratch in C++ with no physics libraries, implementing Newton-Euler rigid-body dynamics with Eigen3
- Replaced Euler with RK4 integration after the simulation diverged; at a 20× larger timestep, RK4 still tracked the fine-step reference trajectory while Euler did not
- Designed a headroom-based motor mixer that scales down attitude commands under actuator saturation to protect collective thrust, eliminating an uncommanded-climb failure
- Built a thrust-vector attitude controller using cross-product attitude error instead of Euler angles, keeping flight stable at high tilt where the angle-based controller broke down

Officer, **LUNAR Undergraduate Council, UT Aerospace** **November 2024 - Present**

- Re-established the department's undergraduate advisory council post-COVID and launched the inaugural Student-Faculty Research Mixer, coordinating faculty to connect undergraduates with research opportunities
- Helped launch an alumni mentorship program (integrated with Cockrell's CAMP), recruiting 10+ senior aerospace/defense alumni (including NASA and Lockheed Orion GNC engineers) as mentors.

SKILLS & ACCOMPLISHMENTS

- **Technical Skills:** Python, C++, MATLAB, ROS 2, Gazebo, OOP, GitHub, SolidWorks, FEA, Soldering
- **Languages:** Chinese - fluent
- **Volunteering:** GivePulse, Storehouse of Collin County, Kids Against Hunger, ABC English Learning